

MICHAEL KINYUNYU

DATA ENGINEER

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PROFESSIONAL SUMMARY

Results-driven Data Engineer with expertise in building scalable ELT/ETL pipelines, modern data stack architectures (dbt, Snowflake, Databricks, Python, AWS), and foundational data layer. Proven track record in establishing medallion architectures, implementing rigorous data quality controls, data contracts, and documenting clear, schema-enforced upstream data models. Experienced in utilizing modern orchestration tools, API integrations, and *AI-powered engineering tools GitHub Copilot* to "*AI-powered engineering tools **such as** GitHub Copilot*"

TECHNICAL SKILLS

- **Data Transformation & Modeling:** dbt, Advanced SQL (CTEs, Complex Joins, Window Functions), Data Modeling, PySpark, Medallion Architecture (Bronze/Silver/Gold)
- **Languages & Frameworks:** Python (Pandas, FastAPI), SQL, PySpark, REST APIs, API Integrations, CDC Workflows
- **Modern Data Stack & Cloud:** Snowflake, Databricks, AWS (S3, Glue, Athena), Apache Kafka, Apache Airflow, Docker
- **Observability & AI Tooling:** AI Agents / LLM Data Assistants (Claude, Copilot), Data Quality & Observability, Tableau, QuickSight, Grafana
- **Engineering Best Practices:** Data Governance, Data Contracts, Version Control (Git), CI/CD, Structured Thinking, Technical Documentation
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PROJECTS

1.Snowflake Analytics Warehouse & dbt Transformation Engine

- **Tech Stack:** dbt, Snowflake, Python, SQL, Git | [github](#)
- Designed and optimized a foundational cloud data warehouse pipeline on Snowflake utilizing dbt best practices for schema-enforced data modeling, lineage tracking, and transformations.
- Implemented extensive dbt testing, schema validation, and observability checks to establish data contracts, eliminate raw data inconsistencies, and ensure highly reliable data products.
- Authored clean, contract-compliant dbt models and comprehensive documentation (dbt docs) to allow analysts, BI teams, and stakeholders to clarify business logic

2.OmniStream Analytics – Multi-Layered E-Commerce Growth Pipeline

- **Tech Stack:** Databricks, PySpark, AWS S3, Python, Streaming/Batch Pipelines | [github link](#)
- Designed and scaled a robust Bronze → Silver → Gold medallion data architecture to ingest raw transactional and event-driven data workflows from AWS S3
- Developed automated data-quality controls and monitoring mechanisms to clean messy data, reconcile schema types, and enforce compliance across high-volume pipelines
- Delivered analytics-ready upstream Gold-layer models optimized for domain teams to evaluate purchasing behavior and business metrics
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3.Autonomous Agentic-RAG Framework (Tanzanian Judiciary)

Tech Stack: Python, RAG, FastAPI, Multi-step Reasoning Agents, Vector Databases | [reasearch journal](#)

- Architected and deployed an AI-driven agentic system capable of parsing, retrieving, and answering complex statutory and structured data questions for non-technical stakeholders.
- Designed natural language interfaces to democratize data access, allowing cross-functional teams to ask automated questions without requiring direct SQL intervention.

4. Healthcare Cancer Data Intelligence Pipeline

- **Tech Stack:** AWS S3, AWS Glue, PySpark, Amazon Athena, Amazon QuickSight | [Github link](#)
- Streamlined REST API ingestion workflows and developed automated PySpark ETL scripts on AWS to clean, normalize, and enforce schemas on messy metadata collections
- Designed curated, partition-optimized Parquet datasets and configured strict catalog access policies to support data governance and self-service querying

EXPERIENCE

Software Engineering Intern

Aug 2025 – Oct 2025

Tanzania Telecommunication Corporation Limited

Iringa, Tanzania

- Developed SQL queries supporting operational reporting and production database activities.
- Built Python scripts to automate recurring system-maintenance tasks and reduce manual operations.
- Generated operational reports using Crystal Reports and Tableau to support business decision-making.
- Participated in software development activities spanning design, implementation, testing, documentation, and deployment.

Software Engineering Intern

Jul 2024 – Sep 2024

TEKU

Mbeya, Tanzania

- Developed SQL queries for application and operational data requirements.
- Built web interfaces using HTML, CSS, and JavaScript.
- Automated recurring operational tasks using Python.
- Investigated and resolved application functionality and debugging issue

EDUCATION

Bachelor of Software Engineering

October 2023

Ruaha Catholic University

Iringa, Tanzania